Find a Real Problem

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Find a Real Problem before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Find a Real Problem trace and record the first step you can explain without help.

Use W2: Identify claims beyond the **evidence** for A problem worth solving. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Find a Real Problem explanation against the supplied facts before discussing it.

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the **outcome**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Find a Real Problem target to a trace and a justified conclusion.

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Find a Real Problem case and show the deciding evidence.

Shared markers are often lost because tray labels are unclear.

State the **need** before selecting AI.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Find a Real Problem adds to Helpful Classroom Sorter and keep the evidence.

Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Write the problem without prescribing a tool

> Write the problem without prescribing a tool. Apply the supplied Find a Real Problem facts and keep your reasoning.

Use the ten-request activity log to write a **problem statement** with a **user**, setting and observed difficulty. Explain why “Build an AI app” is not yet a **problem statement**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Identify claims beyond the evidence

> Identify claims beyond the evidence. Apply the supplied Find a Real Problem facts and keep your reasoning.

A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

> **My reasoning and evidence**

W3 Choose information that serves the outcome

> Choose information that serves the outcome. Apply the supplied Find a Real Problem facts and keep your reasoning.

The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant **input**s from minutes available, quiet required, pupil name and home address.

State the **output** and a simpler alternative to a recommender. Explain one boundary.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Frame a different problem independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Frame a different problem independently. Apply the supplied Find a Real Problem facts and keep your reasoning.

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Measure help as well as avoided errors

> Measure help as well as avoided errors. Apply the supplied Find a Real Problem facts and keep your reasoning.

For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure.

Name two results to record on test requests and explain why the original three-of-eight **observation** is not already an evaluation of the new tool.

> **My reasoning and evidence**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

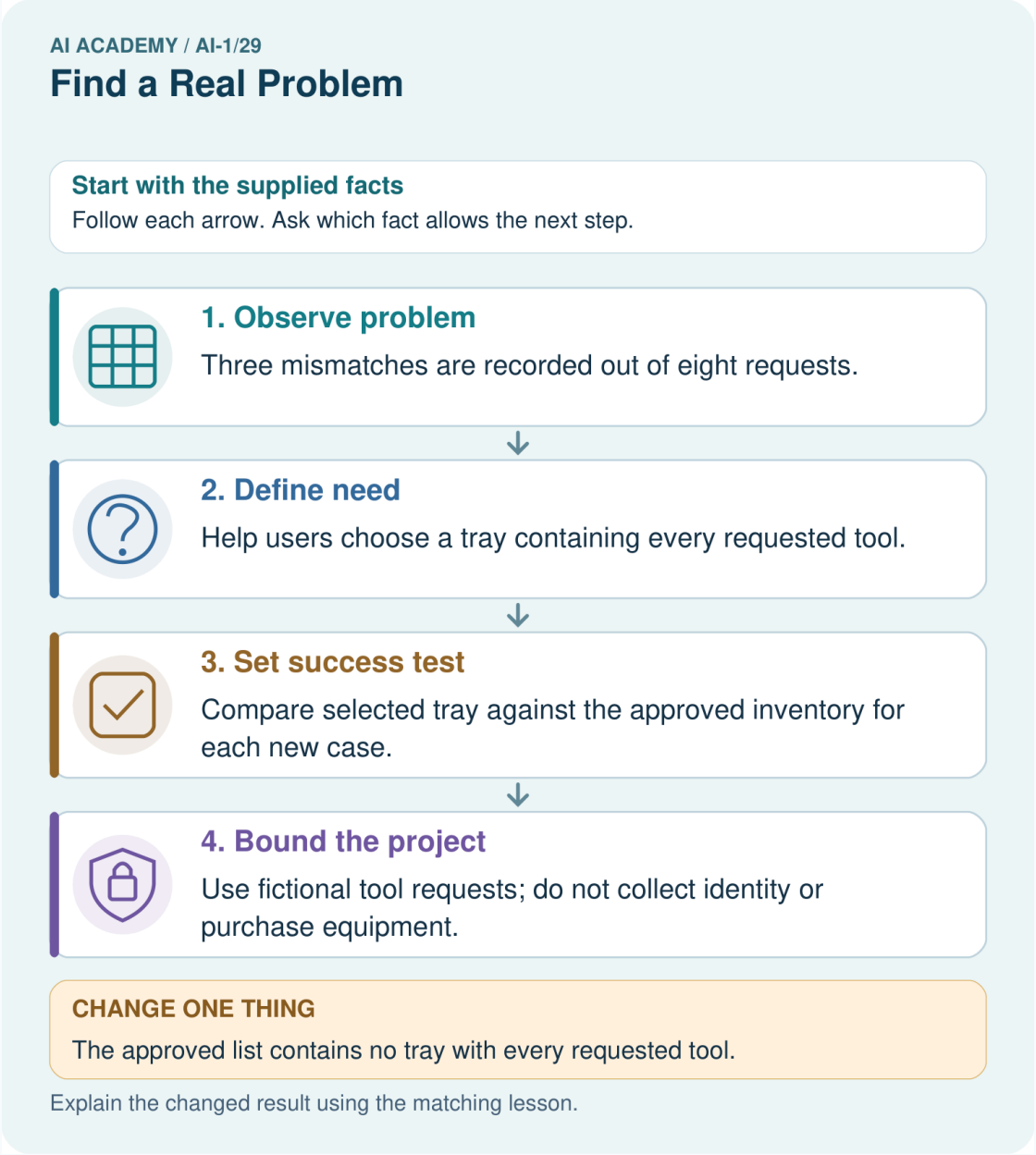
Pictorial case practice: Find a Real Problem

> A fictional class loses time finding the correct craft tray. Explain the labelled step before checking the changed condition.

A fictional class loses time finding the correct craft tray.

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a one-sentence project goal and say what **evidence** would count as a failure.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional class loses time finding the correct craft tray.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.

1. Bound the project

2. Observe problem

3. Set success test

4. Define **need**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete **need**.

A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The finder should report no suitable tray rather than guess.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Repair a missing building block

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two **need** separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

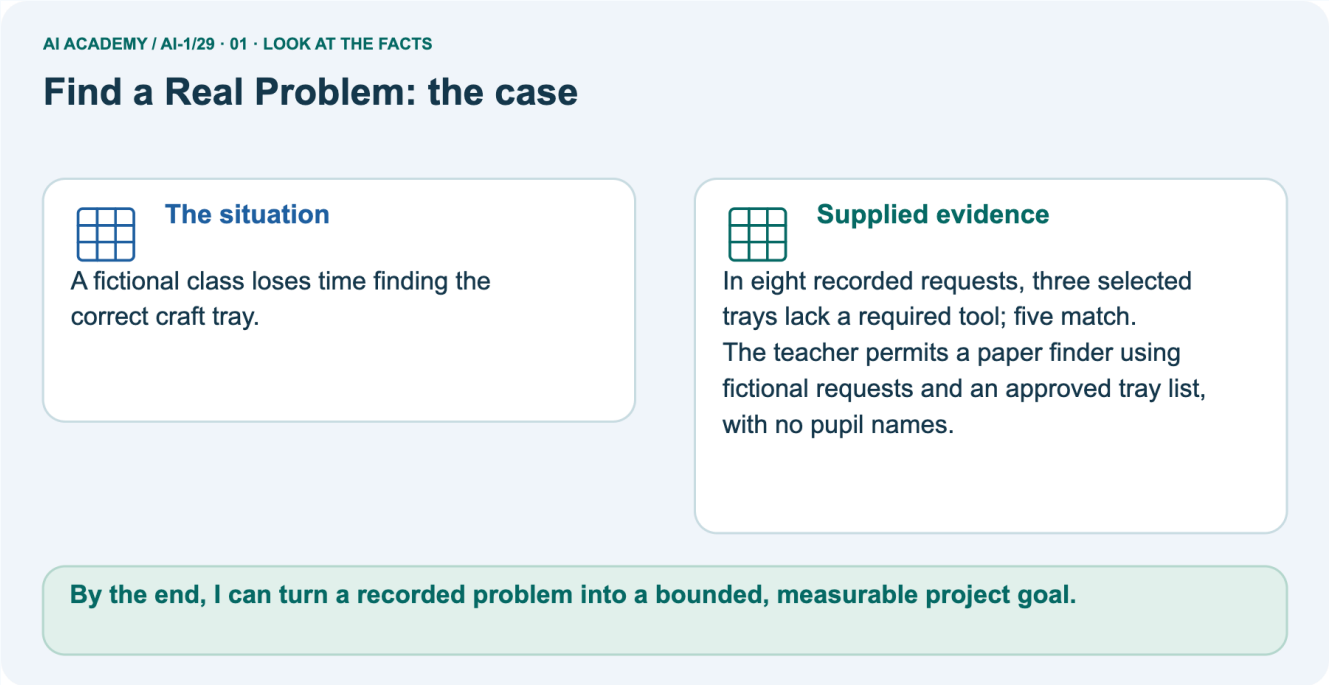
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Find a Real Problem picture from its labels, then record a relevant limit.

01. Find a Real Problem: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

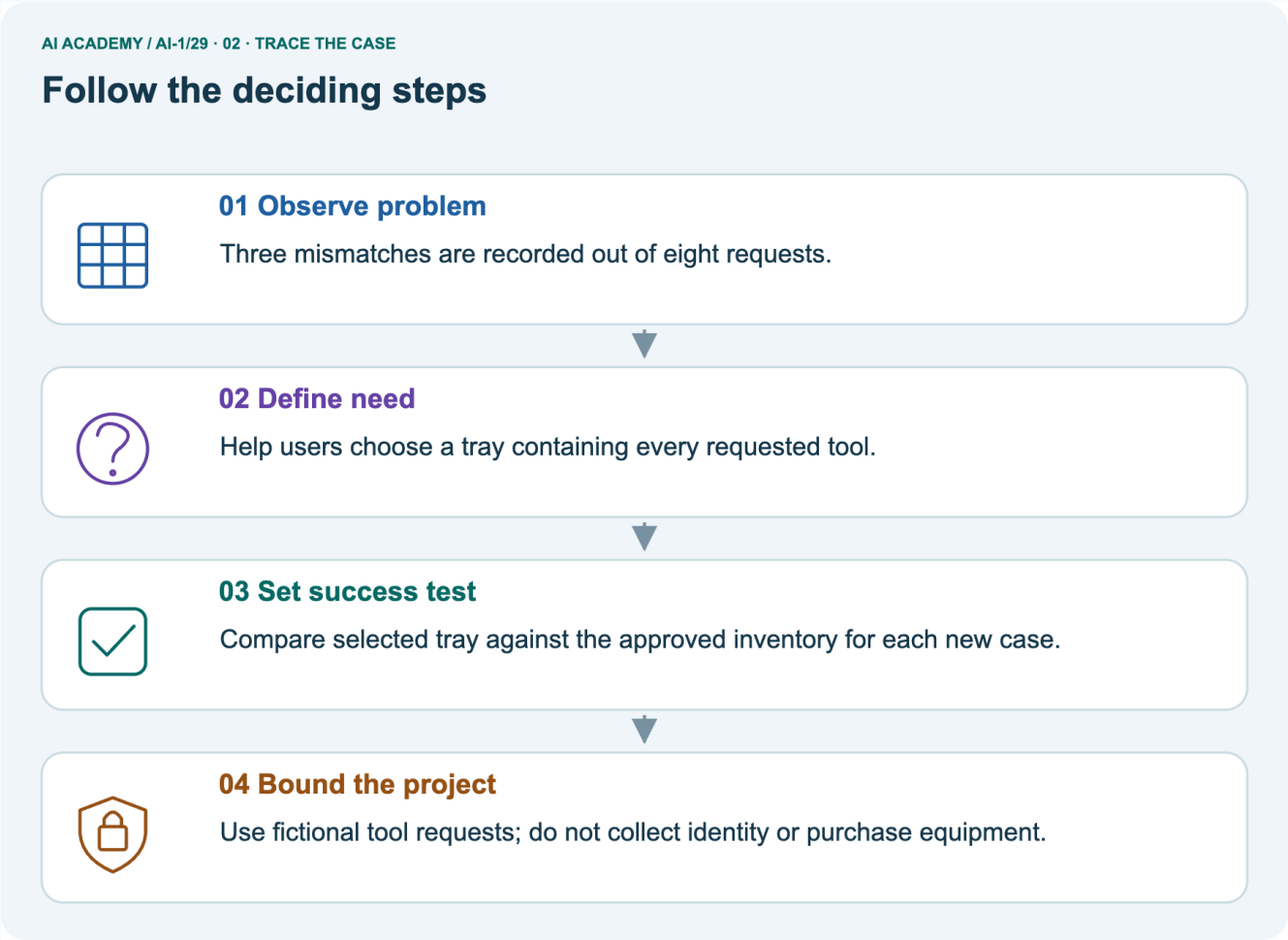
A fictional class loses time finding the correct craft tray. Supplied facts: In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Observe problem: Three mismatches are recorded out of eight requests. → Define **need**: Help **user**s choose a tray containing every requested tool. → Set success test: Compare selected tray against the approved inventory for each new case. → Bound the project: Use fictional tool requests; do not collect identity or purchase equipment.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. From an observed difficulty to a testable goal



Follow the labelled picture with the matching explanation. Open the vector diagram

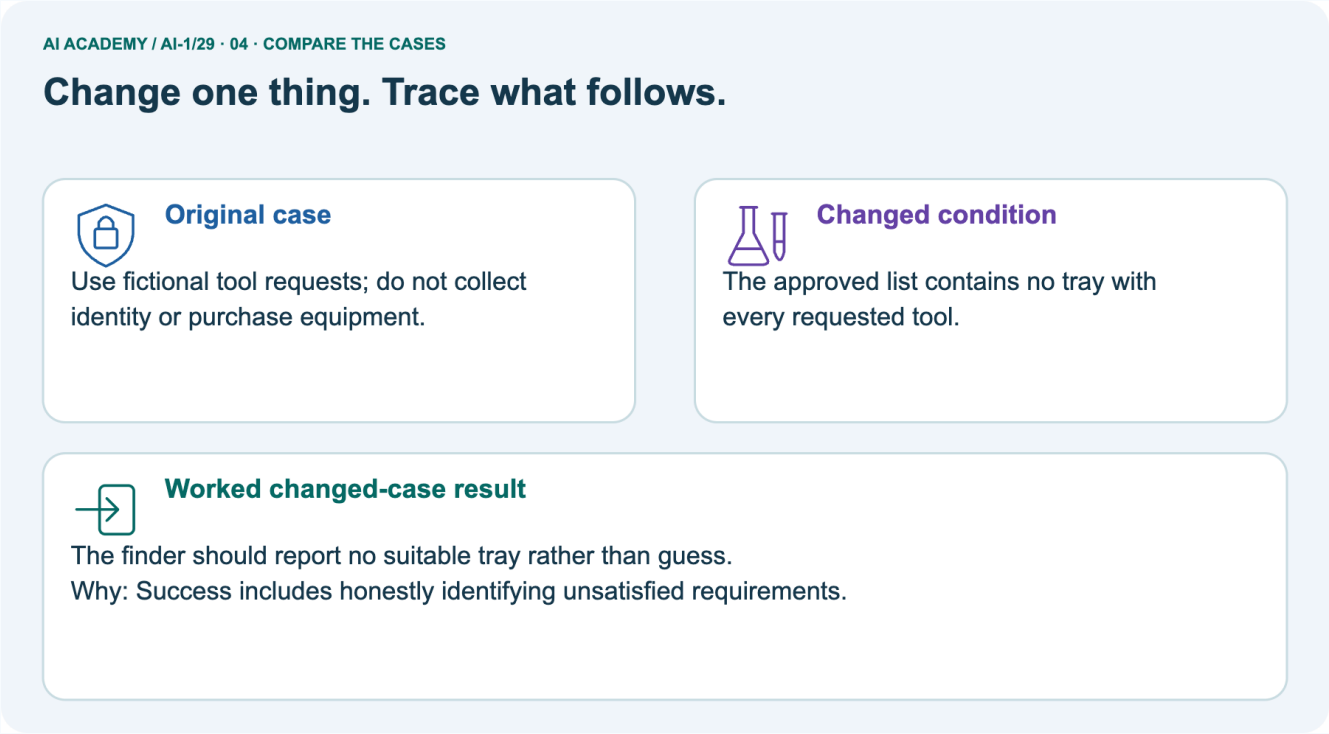
From an observed difficulty to a testable goal. 4 of 10 choices mismatch | Some re-selection occurred | Cause is not established. **Need** a compatible choice | Check time and conditions | **User** can still decline.

Paper suggestion **process** | Small permitted prototype | Benefit **need**s a comparison. Measure fit and coverage | Observe useful **outcome**s | Do not hide no-result cases

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

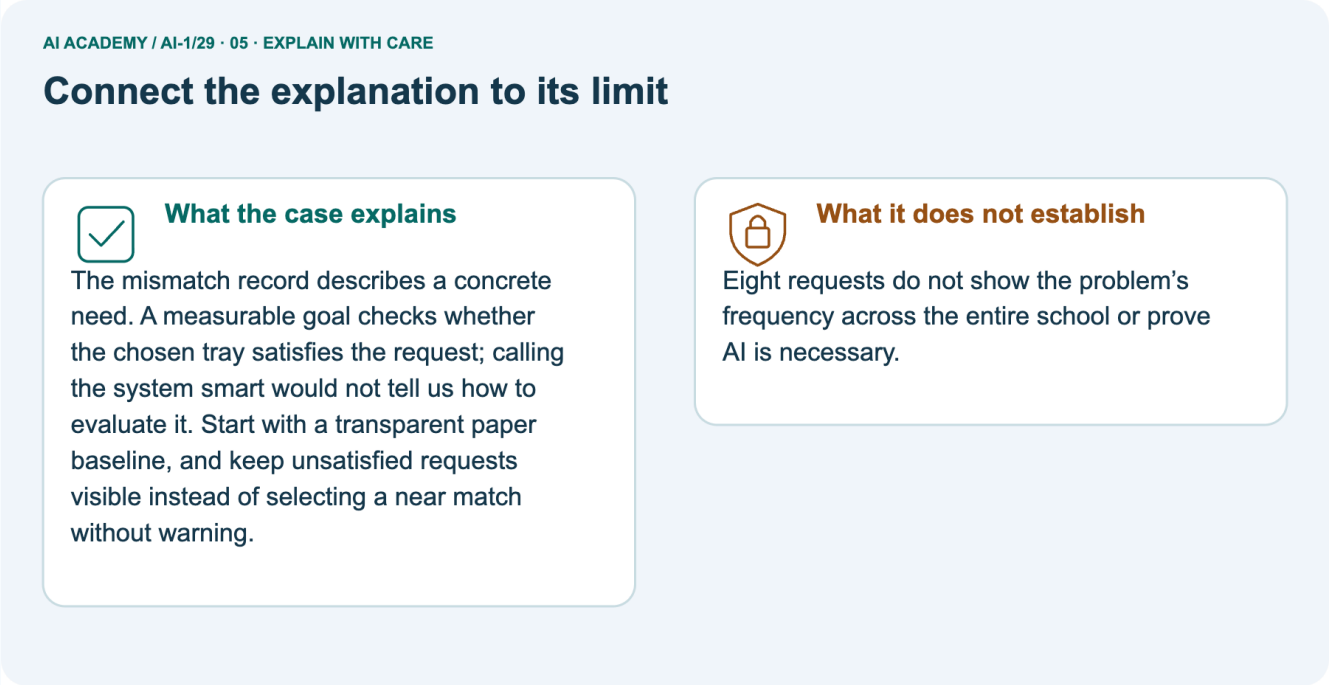
Original case: Use fictional tool requests; do not collect identity or purchase equipment. Changed condition: The approved list contains no tray with every requested tool.

Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

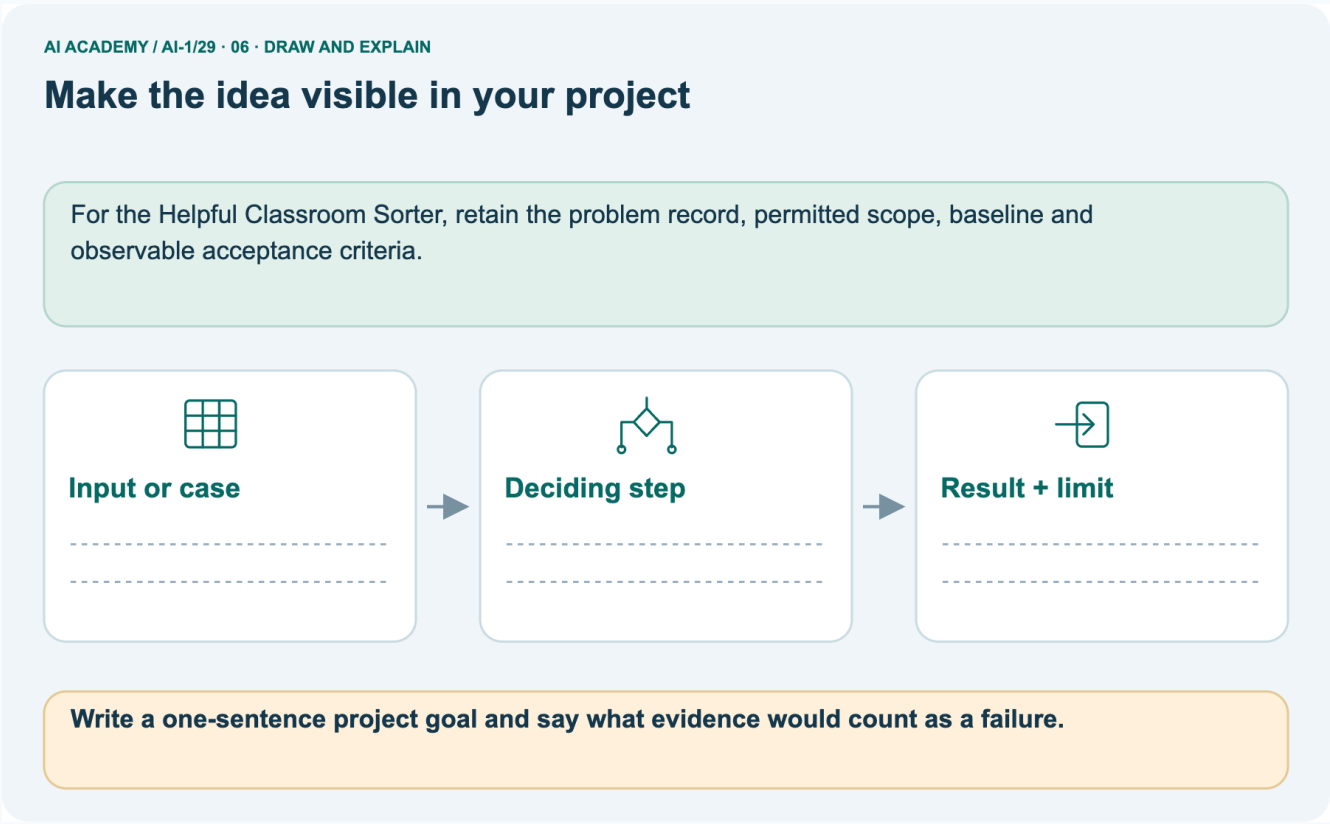
Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Limit: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria. Draw your **input**, deciding step, result and limit.

Supported question: Write a one-sentence project goal and say what **evidence** would count as a failure.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.